

SOC AUTOMATION PROJECT

Engineering Setup — User Manual

Project 1 of 2 | Infrastructure & Configuration Guide

Component	Tool	Version / Tier	Cost
SIEM	Elastic Cloud (Kibana)	9.x — Free 14-day trial	\$0
Log Shipper	Winlogbeat	9.3.3 — Free	\$0
Automation Engine	N8N Cloud	Latest — Free 14-day trial	\$0
AI Analysis	Groq AI	Llama 3.3 70B — Free tier	\$0
Threat Intelligence	AbuseIPDB	API v2 — Free tier	\$0
Notifications	Slack	Free tier	\$0
Hypervisor	VirtualBox	Latest — Free	\$0
Victim Machine	Windows 10 VM	Pro — Free ISO	\$0
Attack Machine	Kali Linux VM	Latest — Free	\$0
TOTAL COST			\$0.00

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1. Project Overview & Architecture

This engineering setup guide covers how to build the complete SOC Automation infrastructure from scratch. By the end of this guide you will have a fully working automated security operations pipeline that detects real attacks and delivers AI-analyzed alerts to Slack in under 90 seconds.

1.1 Complete Architecture

Layer	Component	Purpose
1 — Log Source	Windows 10 VM (192.168.56.102)	Generates Windows Event Logs from user activity and attacks
2 — Log Shipper	Winlogbeat Agent	Reads Windows Event Logs and ships to Elastic Cloud in real time
3 — SIEM	Elastic Cloud (Kibana)	Stores, indexes and analyzes logs — detection rules fire alerts
4 — Automation	N8N Cloud Workflow	Receives webhook, orchestrates AI analysis and notification
5 — AI Analysis	Groq AI — Llama 3.3 70B	Analyzes alert and maps to MITRE ATT&CK technique
6 — Threat Intel	AbuseIPDB API	Enriches IP reputation — abuse score, country, ISP, Tor status
7 — Notification	Slack #alerts Channel	Delivers professional formatted alert to SOC analyst
Attack Machine	Kali Linux VM (192.168.56.101)	Used to simulate attacks against Windows 10 VM

1.2 Network Design

Machine	IP Address	Network Type	Role
Windows 10 VM	192.168.56.102	VirtualBox Host-Only	Victim / Log Source
Kali Linux VM	192.168.56.101	VirtualBox Host-Only	Attacker / Simulator
Elastic Cloud	Cloud (Google Cloud)	Internet	SIEM
N8N Cloud	Cloud (n8n.io)	Internet	Automation Engine

1.3 Data Flow

Attack happens on Windows 10 VM

- Windows Event Log generated (Event IDs 4625, 4688, 4720 etc.)
- Winlogbeat ships log to Elastic Cloud within seconds
- Elastic detection rule matches — alert fires
- Elastic webhook sends JSON payload to N8N
- N8N calls Groq AI API — returns structured analysis
- N8N calls AbuseIPDB API — returns IP reputation
- N8N sends formatted alert to Slack #alerts
- SOC Analyst receives complete alert in under 90 seconds

2. Prerequisites & Hardware Requirements

2.1 Host Machine Requirements

Component	Minimum	Recommended
RAM	16 GB	32 GB
Storage	100 GB free	200 GB free
CPU Cores	4 cores	8 cores
Internet	Stable broadband	Stable broadband
Host OS	Windows 10 / 11	Windows 10 / 11

2.2 Check Virtualization is Enabled

Before creating VMs verify Intel VT-x or AMD-V is enabled:

1. Press Ctrl + Shift + Esc to open Task Manager
2. Click Performance tab
3. Click CPU on left panel
4. Look for Virtualization on the right — must show Enabled

NOTE: If Disabled — restart PC, press F2/Delete to enter BIOS, find Virtualization Technology and set to Enabled

2.3 Software to Download Before Starting

Software	Download URL	Purpose
VirtualBox	virtualbox.org/wiki/Downloads	Run virtual machines
Windows 10 ISO	microsoft.com/software-download/windows10	Windows victim VM
Kali Linux VM	kali.org/get-kali (Virtual Machines)	Attack simulation
Winlogbeat 9.3.3	elastic.co/downloads/beats/winlogbeat	Log shipping agent
7-Zip	7-zip.org	Extract Kali 7z file

3. Account Signups — All Platforms

Create all accounts before starting. Save every credential immediately in a notepad file.

IMPORTANT: Save all credentials before closing any signup page — some passwords are shown only once

3.1 Elastic Cloud (SIEM)

URL: cloud.elastic.co

5. Click Start free trial
6. Use yopmail.com for a temporary email that lasts 8 days
7. Verify email and complete signup
8. Create deployment — Name: SOC-Automation, Region: Asia Pacific (Google Cloud)
9. CRITICAL: When deployment finishes a popup shows your credentials
10. Immediately copy and save: Elasticsearch URL, Kibana URL, Username: elastic, Password
11. You will NEVER see this password again after closing the popup

3.2 N8N Cloud (Automation)

URL: n8n.io

12. Click Get started for free
13. Sign up with email and verify
14. Complete onboarding — skip all paid feature prompts
15. Your workspace URL format: yourname.app.n8n.cloud
16. Save this URL in your credentials notepad

3.3 Groq AI (AI Analysis)

URL: console.groq.com

17. Click Sign Up — use GitHub or email
18. After login click API Keys on left sidebar
19. Click Create API Key
20. Name: SOC-Automation-Key
21. Copy and save the key immediately — starts with gsk_
22. This key is shown only once

3.4 AbuseIPDB (Threat Intelligence)

URL: abuseipdb.com

23. Click Sign Up top right
24. Create account and verify email
25. Go to API section in top navigation
26. Click Create Key — Name: SOC-Project
27. Copy and save the API key

3.5 Slack (Notifications)

URL: slack.com

28. Click Get Started
29. Create new workspace — Name: SOC-Automation
30. Create a channel named: alerts (lowercase, set to Public)
31. Save the workspace URL

3.6 Credentials Template — Save This Now

Open Notepad and save with your actual values:

```
===== SOC AUTOMATION CREDENTIALS =====

ELASTIC CLOUD
Kibana URL      : https://xxxxx.kb.us-central1.gcp.cloud.es.io
Elastic URL     : https://xxxxx.es.us-central1.gcp.cloud.es.io
Username        : elastic
Password        : (copied from deployment popup)

N8N CLOUD
Workspace URL   : https://yourname.app.n8n.cloud

GROQ AI
API Key         : gsk_XXXXXXXXXXXXXXXXXXXXX

ABUSEIPDB
API Key         : XXXXXXXXXXXXXXXXXXXXXXXX

SLACK
Workspace       : SOC-Automation
Channel         : #alerts

VM NETWORK
Windows 10 VM   : 192.168.56.102
Kali Linux VM   : 192.168.56.101
```

4. Virtual Machine Setup (VirtualBox)

4.1 Install VirtualBox

32. Go to [virtualbox.org/wiki/Downloads](https://www.virtualbox.org/wiki/Downloads)
33. Click Windows hosts to download the installer
34. Run the installer — accept all defaults
35. Restart your computer after installation completes

4.2 Create Windows 10 VM

36. Download Windows 10 ISO from [microsoft.com/software-download/windows10](https://www.microsoft.com/software-download/windows10)
37. Select Create Windows 10 installation media then choose ISO file
38. Save the ISO to your Downloads folder — takes 10-15 minutes
39. Open VirtualBox and click New button top left
40. Configure the VM with these settings:
 - Name: Windows-10-SOC
 - Type: Microsoft Windows
 - Version: Windows 10 (64-bit)
 - Memory: 4096 MB minimum (8192 MB recommended)
 - Hard Disk: Create new — 60 GB — VDI format — Dynamically allocated
41. Before starting — go to Settings > Storage > Controller IDE > Empty > click disc icon > Choose ISO file > select your Windows 10 ISO
42. Start the VM and complete Windows installation:
 - Select Windows 10 Pro edition
 - Select I don't have a product key
 - Choose Custom installation
 - Username: windows
 - Set a password you will remember

Configure Host-Only Network on Windows 10 VM

43. With VM powered off — click Settings in VirtualBox
44. Click Network on left
45. Adapter 1: Change Attached to from NAT to Host-Only Adapter
46. Name: VirtualBox Host-Only Ethernet Adapter
47. Click OK

Get Windows 10 VM IP Address

48. Start Windows 10 VM and login
49. Open Command Prompt — Start > cmd
50. Type: ipconfig and press Enter

51. Note the IPv4 Address — example: 192.168.56.102
52. Save this in your credentials notepad

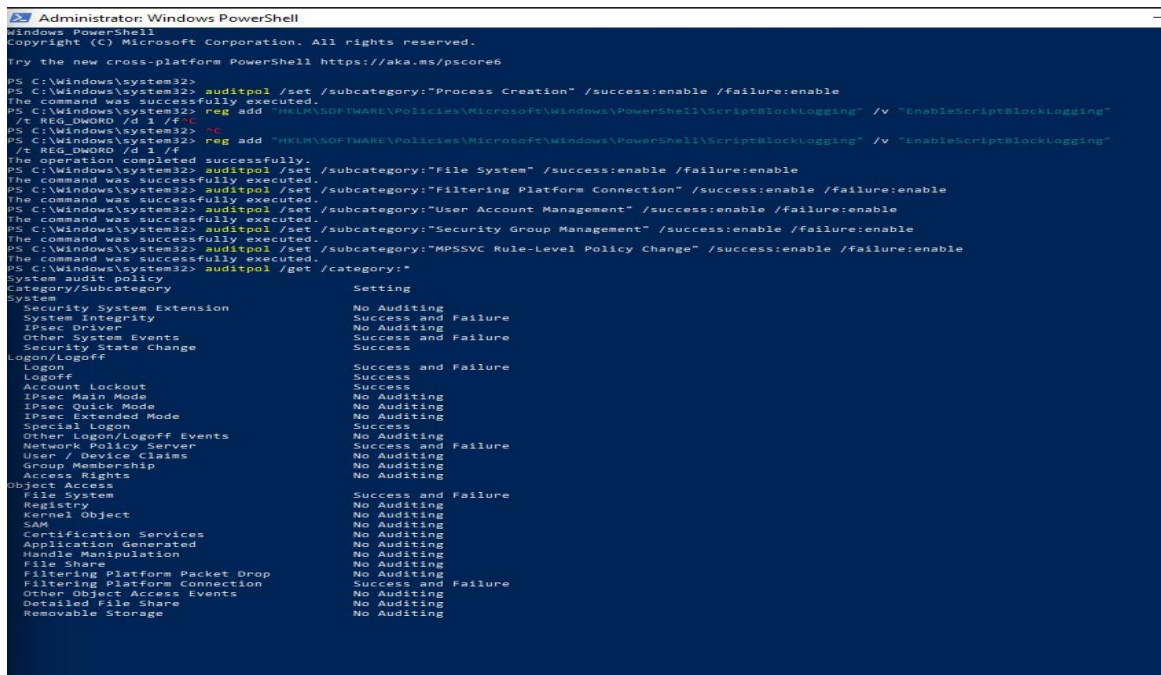
5. Windows 10 VM Configuration

These configurations are required to capture all security events that our detection rules need. Open PowerShell as Administrator for all commands.

5.1 Enable Windows Audit Policies

These policies tell Windows to log specific security events:

```
auditpol /set /subcategory:"Process Creation" /success:enable /failure:enable
auditpol /set /subcategory:"File System" /success:enable /failure:enable
auditpol /set /subcategory:"Filtering Platform Connection" /success:enable /failure:enable
auditpol /set /subcategory:"User Account Management" /success:enable /failure:enable
auditpol /set /subcategory:"Security Group Management" /success:enable /failure:enable
auditpol /set /subcategory:"MPSSVC Rule-Level Policy Change" /success:enable /failure:enable
auditpol /set /subcategory:"Credential Validation" /success:enable /failure:enable
auditpol /set /subcategory:"Handle Manipulation" /success:enable /failure:enable
```



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> auditpol /set /subcategory:"Process Creation" /success:enable /failure:enable
The command was successfully executed.
PS C:\Windows\system32> reg add "HKLM\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging" /v "EnableScriptBlockLogging" /t REG_DWORD /d 1 /f
PS C:\Windows\system32> reg add "HKLM\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging" /v "EnableScriptBlockLogging" /t REG_DWORD /d 1 /f
The operation completed successfully.
PS C:\Windows\system32> auditpol /set /subcategory:"File System" /success:enable /failure:enable
The command was successfully executed.
PS C:\Windows\system32> auditpol /set /subcategory:"Filtering Platform Connection" /success:enable /failure:enable
The command was successfully executed.
PS C:\Windows\system32> auditpol /set /subcategory:"User Account Management" /success:enable /failure:enable
The command was successfully executed.
PS C:\Windows\system32> auditpol /set /subcategory:"Security Group Management" /success:enable /failure:enable
The command was successfully executed.
PS C:\Windows\system32> auditpol /set /subcategory:"MPSSVC Rule-Level Policy Change" /success:enable /failure:enable
The command was successfully executed.
PS C:\Windows\system32> auditpol /get /category:*
System audit policy
Category/Subcategory      Setting
-----
System
Security System Extension  No Auditing
System Integrity           Success and Failure
IPsec Driver               No Auditing
Other System Events        Success and Failure
Security State Change      Success
Logon/Logoff
Logon                      Success and Failure
Logoff                     Success
Account Lockout            Success
IPsec Main Mode            No Auditing
IPsec Quick Mode           No Auditing
IPsec Extended Mode        No Auditing
Special Logon              Success
Other Logon/Logoff Events  No Auditing
Network Policy Server      Success and Failure
User / Device Claims       No Auditing
Group Membership           No Auditing
Access Rights              No Auditing
Object Access
File System                Success and Failure
Registry                  No Auditing
Kernel Object              No Auditing
SAM                        No Auditing
Certification Services     No Auditing
Application Generated      No Auditing
Handle Manipulation         No Auditing
File Share                 No Auditing
Filtering Platform Packet Drop No Auditing
Filtering Platform Connection Success and Failure
Other Object Access Events No Auditing
Detailed File Share        No Auditing
Removable Storage          No Auditing
```

Windows audit policies enabled — all subcategories set to Success and Failure

5.2 Enable PowerShell Script Block Logging

Required to capture Event ID 4104 — PowerShell script content:

```
New-Item -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging" -Force
Set-ItemProperty -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging" -Name "EnableScriptBlockLogging" -Value 1 -Force
```

5.3 Enable Process Command Line Logging

Required to capture the full command line in Event ID 4688:

```
reg add "HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\Audit" /v "ProcessCreationIncludeCmdLine_Enabled" /t REG_DWORD /d 1 /f
```

5.4 Verify Audit Policies

```
auditpol /get /category:"Account Logon"
auditpol /get /category:"Account Management"
```

Look for Success and Failure next to each policy line.

6. Kali Linux VM Setup

6.1 Import Kali Linux

- 53. Go to kali.org/get-kali and click Virtual Machines tab
- 54. Download VirtualBox 64-bit version — it is a 7z compressed file
- 55. Extract using 7-Zip — right click > 7-Zip > Extract Here
- 56. Double click the .vbox file inside the extracted folder
- 57. VirtualBox automatically imports the VM

6.2 Configure Network

- 58. With Kali VM powered off click Settings in VirtualBox
- 59. Click Network > Adapter 1
- 60. Change Attached to from NAT to Host-Only Adapter
- 61. Same adapter as Windows 10 VM — VirtualBox Host-Only Ethernet Adapter
- 62. Click OK

6.3 Login and Verify Network

- 63. Start Kali VM
- 64. Default credentials: username kali, password kali
- 65. Open terminal and run:
- 66. Note the IP address — example: 192.168.56.101
- 67. Test connectivity to Windows VM:

```
ip a
```

```
ping 192.168.56.102
```

```
valid_lft forever preferred_lft forever

(kali㉿kali)-[~]
$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=128 time=3.58 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=128 time=3.17 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=128 time=3.23 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=128 time=4.79 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=128 time=2.98 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=128 time=2.17 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=128 time=3.05 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=128 time=2.72 ms
64 bytes from 192.168.56.102: icmp_seq=9 ttl=128 time=2.59 ms

64 bytes from 192.168.56.102: icmp_seq=10 ttl=128 time=2.62 ms
64 bytes from 192.168.56.102: icmp_seq=11 ttl=128 time=2.18 ms
^C
— 192.168.56.102 ping statistics —
11 packets transmitted, 11 received, 0% packet loss, time 10275ms
rtt min/avg/max/mdev = 2.172/3.006/4.788/0.698 ms

(kali㉿kali)-[~]
$
```

Kali Linux terminal showing network connectivity to Windows 10 VM — both VMs on same Host-Only network

7. Winlogbeat Installation & Configuration

Winlogbeat is the log shipping agent that reads Windows Event Logs and forwards them to Elastic Cloud in real time. All steps are done on the Windows 10 VM.

7.1 Download and Extract Winlogbeat

68. On Windows 10 VM open a browser
69. Go to elastic.co/downloads/beats/winlogbeat
70. Download the Windows ZIP version (winlogbeat-9.3.3-windows-x86_64.zip)
71. Save to C:\Users\windows\Downloads\
72. Right click the ZIP > Extract All
73. Extract to: C:\Users\windows\Downloads\Winlogbeat\

7.2 Configure winlogbeat.yml

74. Navigate to the extracted Winlogbeat folder
75. Right click winlogbeat.yml > Open with Notepad
76. Select all with Ctrl+A and delete everything
77. Paste this complete configuration — replace the 3 YOUR_ values:

```
winlogbeat.event_logs:
  - name: Application
    ignore_older: 72h
  - name: System
    ignore_older: 72h
  - name: Security
    ignore_older: 72h
    event_id: 4624, 4625, 4634, 4648, 4688, 4720, 4722,
              4724, 4728, 4732, 4740, 4756, 4950, 4954, 5156
  - name: Microsoft-Windows-PowerShell/Operational
    ignore_older: 72h
    event_id: 4103, 4104, 4105, 4106

output.elasticsearch:
  hosts: ["YOUR_ELASTICSEARCH_URL:443"]
  protocol: "https"
  username: "elastic"
  password: "YOUR_ELASTIC_PASSWORD"

setup.kibana:
  host: "YOUR_KIBANA_URL:443"

logging.level: info
logging.to_files: true
logging.files:
```

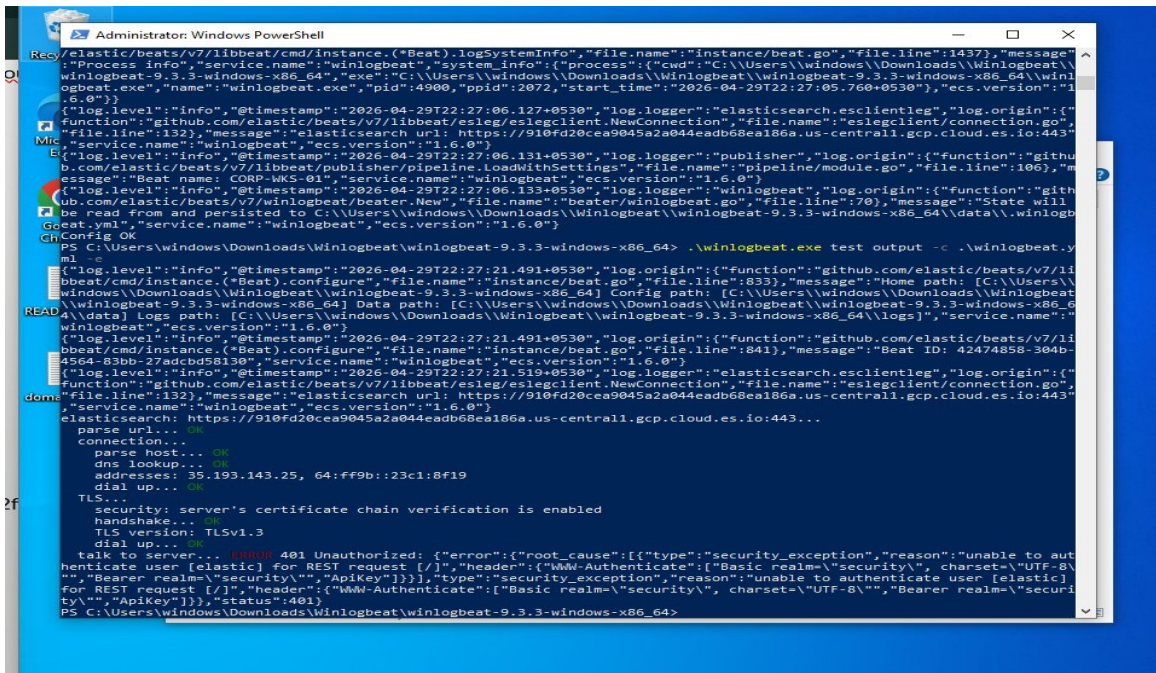
```
path: C:/Users/windows/Downloads/Winlogbeat/winlogbeat-9.3.3-windows-x86_64/
Logs
name: winlogbeat
keepfiles: 7
permissions: 0644
```

NOTE: Replace YOUR_ELASTICSEARCH_URL with your Elasticsearch endpoint URL, YOUR_ELASTIC_PASSWORD with your elastic password, and YOUR_KIBANA_URL with your Kibana URL. All three are in your credentials notepad from Section 3.

7.3 Install Winlogbeat as Windows Service

Open PowerShell as Administrator and run these commands one by one:

```
cd "C:\Users\windows\Downloads\Winlogbeat\winlogbeat-9.3.3-windows-x86_64"
PowerShell -ExecutionPolicy Bypass -File .\install-service-winlogbeat.ps1
Start-Service winlogbeat
Get-Service winlogbeat
```



```
Administrator: Windows PowerShell
PS C:\Users\windows\Downloads\Winlogbeat\winlogbeat-9.3.3-windows-x86_64> .\winlogbeat.exe test output -c .\winlogbeat.yml
{"log.level":"info","@timestamp":"2026-04-29T22:27:06.127+0530","log.logger":"elasticsearch.esclientleg","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).logSystemInfo","file.name":"instance/beat.go","file.line":1437},"message":{"process_info":{"service.name":"winlogbeat"},"system_info":{"cwd":"C:\\Users\\windows\\Downloads\\Winlogbeat\\winlogbeat-9.3.3-windows-x86_64","exe":"C:\\Users\\windows\\Downloads\\Winlogbeat\\winlogbeat-9.3.3-windows-x86_64\\winlogbeat.exe","name":"winlogbeat.exe","pid":4900,"ppid":2072,"start_time":"2026-04-29T22:27:05.760+0530"},"ecs.version":"1.6.0"}}
{"log.level":"info","@timestamp":"2026-04-29T22:27:06.131+0530","log.logger":"publisher","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/esleg/eslegclient.NewConnection","file.name":"eslegclient/connection.go","file.line":132},"message":{"elasticsearch_url":"https://910fd20cea9045a2a044eadb68ea186a.us-central1.gcp.cloud.es.io:443"},"service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2026-04-29T22:27:06.133+0530","log.logger":"winlogbeat","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/publisher/pipeline.LoadWithSettings","file.name":"pipeline/module.go","file.line":106},"message":{"Beat name: CORP-WKS-01","service.name":"winlogbeat","ecs.version":"1.6.0"}}
{"log.level":"info","@timestamp":"2026-04-29T22:27:06.133+0530","log.logger":"winlogbeat","log.origin":{"function":"github.com/elastic/beats/v7/winlogbeat/beater.New","file.name":"beater/winlogbeat.go","file.line":70},"message":{"State will be read from and persisted to C:\\Users\\windows\\Downloads\\Winlogbeat\\winlogbeat-9.3.3-windows-x86_64\\data\\winlogbeat.yml"},"service.name":"winlogbeat","ecs.version":"1.6.0"}
PS C:\Users\windows\Downloads\Winlogbeat\winlogbeat-9.3.3-windows-x86_64> .\winlogbeat.exe test output -c .\winlogbeat.yml
{"log.level":"info","@timestamp":"2026-04-29T22:27:21.491+0530","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).configure","file.name":"instance/beat.go","file.line":833},"message":{"Home path: [C:\\Users\\windows\\Downloads\\Winlogbeat\\winlogbeat-9.3.3-windows-x86_64] Config path: [C:\\Users\\windows\\Downloads\\Winlogbeat\\winlogbeat-9.3.3-windows-x86_64\\data] Logs path: [C:\\Users\\windows\\Downloads\\Winlogbeat\\winlogbeat-9.3.3-windows-x86_64\\logs]","service.name":"winlogbeat","ecs.version":"1.6.0"}}
{"log.level":"info","@timestamp":"2026-04-29T22:27:21.519+0530","log.logger":"elasticsearch.esclientleg","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/esleg/eslegclient.NewConnection","file.name":"eslegclient/connection.go","file.line":132},"message":{"elasticsearch_url":"https://910fd20cea9045a2a044eadb68ea186a.us-central1.gcp.cloud.es.io:443"},"service.name":"winlogbeat","ecs.version":"1.6.0"}
elasticsearch: https://910fd20cea9045a2a044eadb68ea186a.us-central1.gcp.cloud.es.io:443...
  parse url... OK
  connection... OK
  parse host... OK
  dns lookup... OK
  addresses: 35.193.143.25, 64:ff9b:23c1:8f19
  dial up... OK
  TLS... OK
    security: server's certificate chain verification is enabled
  handshake... OK
  TLS version: TLSv1.3
  talk to server... ERROR 401 Unauthorized: {"error":{"root_cause":[{"type":"security_exception","reason":"unable to authenticate user [elastic] for REST request [/]","header":{"WWW-Authenticate":["Basic realm=\"security\"","Bearer realm=\"security\"","ApiKey"]}],type":"security_exception","reason":"unable to authenticate user [elastic] for REST request [/]","header":{"WWW-Authenticate":["Basic realm=\"security\"","Bearer realm=\"security\"","ApiKey"]}],status:401}}
PS C:\Users\windows\Downloads\Winlogbeat\winlogbeat-9.3.3-windows-x86_64>
```

PowerShell showing Get-Service winlogbeat with Status: Running — Winlogbeat service is active

7.4 Test Connection to Elastic Cloud

```
.\winlogbeat.exe test output -c .\winlogbeat.yml -e
```

```
Administrator: Windows PowerShell
Recy dns lookup... OK
addresses: 35.193.143.25, 64:ff9b:23c1:8f19
dial up... OK
TLS...
security: server's certificate chain verification is enabled
handshake... OK
TLS version: TLSv1.3
dial up... OK
talk to server... 401 Unauthorized: {"error":{"root_cause":[{"type":"security_exception","reason":"unable to aut
authenticate user [elastic] for REST request [/]","header":{"WWW-Authenticate":["Basic realm=\"security\", charset=\"UTF-8\"
\"\", \"Bearer realm=\"security\", \"ApiKey\"]}],\"type\":\"security_exception\",\"reason\":\"unable to authenticate user [elastic]
for REST request [/]\",\"header":{"WWW-Authenticate":["Basic realm=\"security\", charset=\"UTF-8\"\", \"Bearer realm=\"securi
ty\", \"ApiKey\"]}],\"status\":401}
PS C:\Users\windows\Downloads\Winlogbeat\winlogbeat-9.3.3-windows-x86_64> notepad .\winlogbeat.yml
PS C:\Users\windows\Downloads\Winlogbeat\winlogbeat-9.3.3-windows-x86_64> Stop-Service winlogbeat
GoPS C:\Users\windows\Downloads\Winlogbeat\winlogbeat-9.3.3-windows-x86_64> Start-Service winlogbeat
CHPS C:\Users\windows\Downloads\Winlogbeat\winlogbeat-9.3.3-windows-x86_64> .\winlogbeat.exe test output -c .\winlogbeat.y
ml -e
{"log.level":"info","@timestamp":"2026-04-29T22:33:54.282+0530","log.origin":{"function":"github.com/elastic/beats/v7/li
bbeat/cmd/instance.(*Beat).configure","file.name":"instance/beat.go","file.line":833},"message":"Home path: [C:\\Users\\
windows\\Downloads\\Winlogbeat\\winlogbeat-9.3.3-windows-x86_64] Config path: [C:\\Users\\windows\\Downloads\\Winlogbeat
\\winlogbeat-9.3.3-windows-x86_64] Data path: [C:\\Users\\windows\\Downloads\\Winlogbeat\\winlogbeat-9.3.3-windows-x86_6
4\\data] logs path: [C:\\Users\\windows\\Downloads\\Winlogbeat\\winlogbeat-9.3.3-windows-x86_64\\logs]","service.name":"
winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2026-04-29T22:33:54.286+0530","log.origin":{"function":"github.com/elastic/beats/v7/li
bbeat/cmd/instance.(*Beat).configure","file.name":"instance/beat.go","file.line":841},"message":"Beat ID: 42474858-304b-
4964-83bb-27adcd0b58130","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2026-04-29T22:33:54.327+0530","log.logger":"elasticsearch.esclientleg","log.origin":{"f
unction":"github.com/elastic/beats/v7/libbeat/esleg/eslegclient.NewConnection","file.name":"eslegclient/connection.go","
file.line":132},"message":"elasticsearch url: https://910fd20cea9045a2a044eadb68ea186a.us-central1.gcp.cloud.es.io:443...
dome","service.name":"winlogbeat","ecs.version":"1.6.0"}
elasticsearch: https://910fd20cea9045a2a044eadb68ea186a.us-central1.gcp.cloud.es.io:443...
parse url... OK
connection...
parse host... OK
dns lookup... OK
addresses: 35.193.143.25, 64:ff9b:23c1:8f19
dial up... OK
TLS...
security: server's certificate chain verification is enabled
handshake... OK
TLS version: TLSv1.3
dial up... OK
{"log.level":"info","@timestamp":"2026-04-29T22:33:58.402+0530","log.logger":"elasticsearch.esclientleg","log.origin":{"f
unction":"github.com/elastic/beats/v7/libbeat/esleg/eslegclient.(*Connection).Ping","file.name":"eslegclient/connection
.go","file.line":323},"message":"Attempting to connect to Elasticsearch version 9.3.3 (default)","service.name":"winlog
beat","ecs.version":"1.6.0"}
version: 9.3.3
PS C:\Users\windows\Downloads\Winlogbeat\winlogbeat-9.3.3-windows-x86_64>
```

PowerShell showing winlogbeat test output with talk to server... OK — confirms successful connection to Elastic Cloud

NOTE: If you see 401 Unauthorized — your password in winlogbeat.yml is wrong. Go to cloud.elastic.co and reset the elastic password, then update winlogbeat.yml.

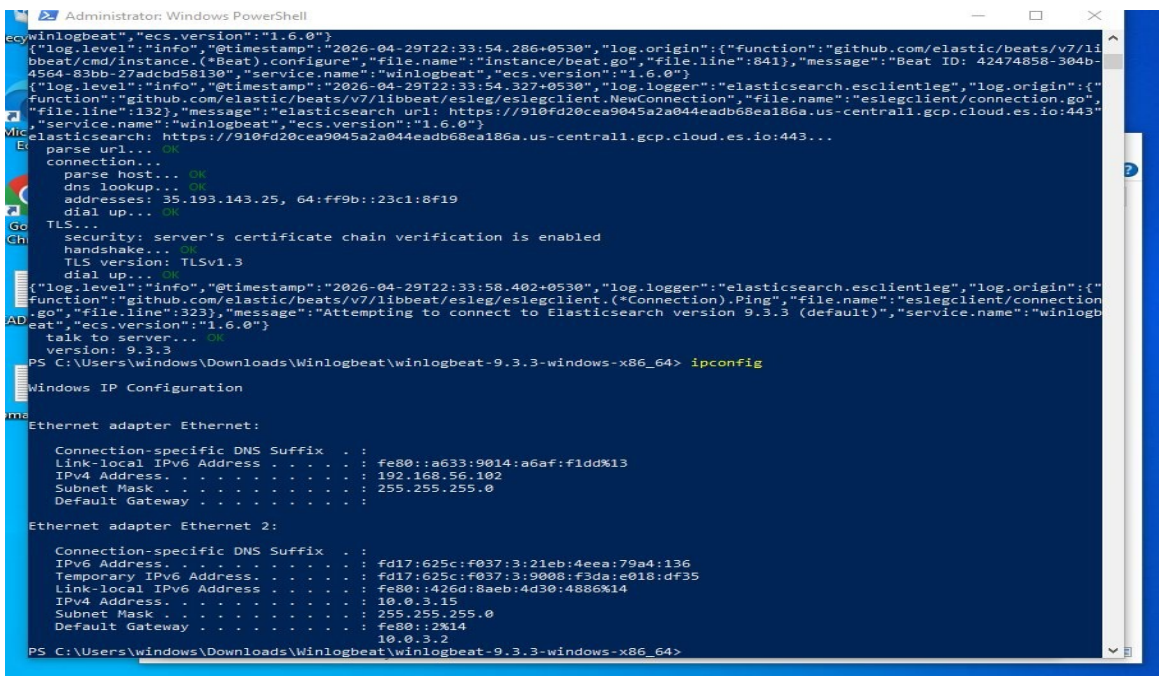
8. Elastic Cloud SIEM Setup

8.1 Access Kibana

78. Go to cloud.elastic.co and login
79. Click your SOC-Automation deployment
80. Click Open Kibana button
81. Login with username: elastic and your saved password

8.2 Verify Logs Are Flowing

82. In Kibana click hamburger menu (three lines) top left
83. Click Discover
84. In search bar type: event.code: "4624"
85. Set time range to Last 15 minutes top right
86. You should see Windows login events from CORP-WKS-01



```
Administrator: Windows PowerShell
C:\Users\windows\Downloads>winlogbeat
{"log.level":"info","@timestamp":"2026-04-29T22:33:54.286+0530","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/instance.(*Beat).configure","file.name":"instance/beat.go","file.line":841},"message":"Beat ID: 42474858-304b-4564-83bb-27adcd858130","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2026-04-29T22:33:54.327+0530","log.logger":"elasticsearch.esclientleg","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/esleg/eslegclient.NewConnection","file.name":"eslegclient/connection.go","file.line":132},"message":"elasticsearch url: https://910fd20cea9045a2a044eadb68ea186a.us-central1.gcp.cloud.es.io:443","service.name":"winlogbeat","ecs.version":"1.6.0"}
elasticsearch: https://910fd20cea9045a2a044eadb68ea186a.us-central1.gcp.cloud.es.io:443...
  parse url... OK
  connection...
  parse host... OK
  dns lookup...
    addresses: 35.193.143.25, 64:ff9b::23c1:8f19
  dial up... OK
  TLS...
    security: server's certificate chain verification is enabled
    handshake... OK
    TLS version: TLSv1.3
  dial up... OK
{"log.level":"info","@timestamp":"2026-04-29T22:33:58.402+0530","log.logger":"elasticsearch.esclientleg","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/esleg/eslegclient.(*Connection).Ping","file.name":"eslegclient/connection.go","file.line":323},"message":"Attempting to connect to Elasticsearch version 9.3.3 (default)","service.name":"winlogbeat","ecs.version":"1.6.0"}
  talk to server... OK
  version: 9.3.3
PS C:\Users\windows\Downloads>Winlogbeat\winlogbeat-9.3.3-windows-x86_64> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::a633:9014:a6af:f1dd%13
    IPv4 Address. . . . . : 192.168.56.102
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : fd17:625c:f037:3:21eb:4eea:79a4:136
    Temporary IPv6 Address. . . . . : fd17:625c:f037:3:9008:f3da:e018:df35
    Link-local IPv6 Address . . . . . : fe80::426d:8aeb:4d30:4886%14
    IPv4 Address. . . . . : 10.0.3.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2%14
    10.0.3.2
PS C:\Users\windows\Downloads>Winlogbeat\winlogbeat-9.3.3-windows-x86_64>
```

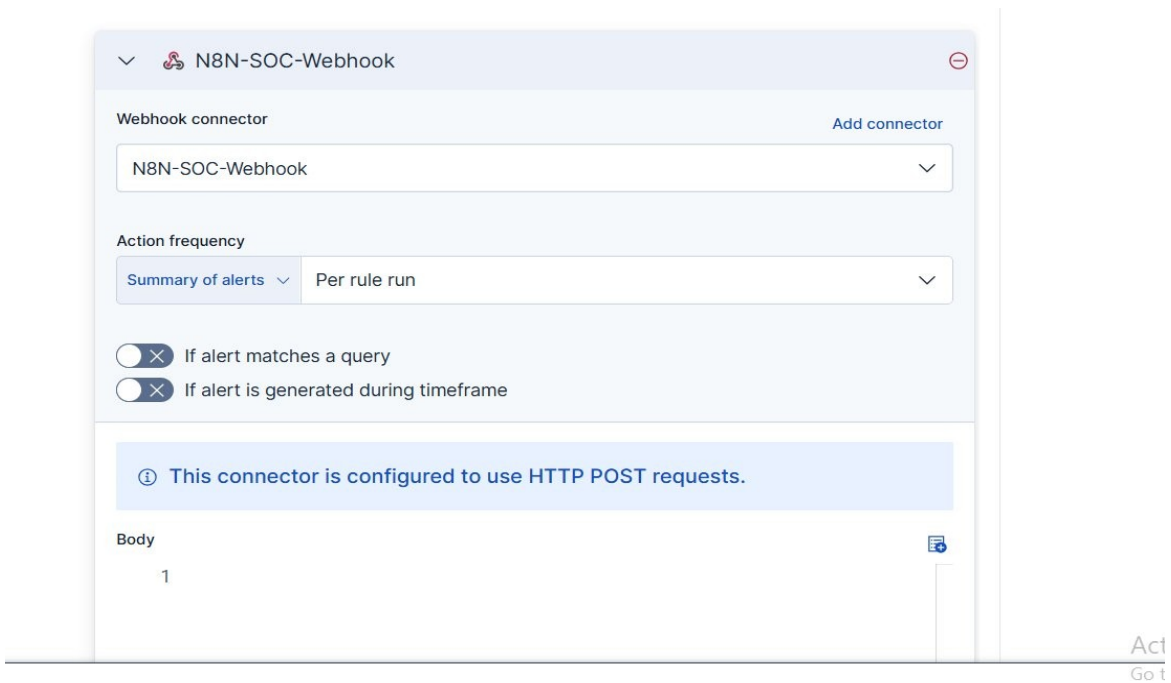
Kibana Discover showing Windows security events flowing in from CORP-WKS-01 — confirms Winlogbeat to Elastic pipeline is working

8.3 Create N8N Webhook Connector

NOTE: Get your N8N Production URL first from Section 10.2, then come back to create this connector.

87. In Kibana click hamburger menu > Stack Management
88. Under Alerts and Insights click Connectors

89. Click Create connector > select Webhook
90. Fill in these details:
 - Connector name: N8N-SOC-Webhook
 - Method: POST
 - URL: Your N8N Production Webhook URL
 - Authentication: None
91. Click Add HTTP header:
 - Key: Content-Type
 - Value: application/json
92. Click Save and Test



Elastic Kibana showing N8N-SOC-Webhook connector saved and ready — POST method configured with Content-Type header

8.4 Webhook Body Template

Use this exact JSON body when adding webhook actions to every detection rule:

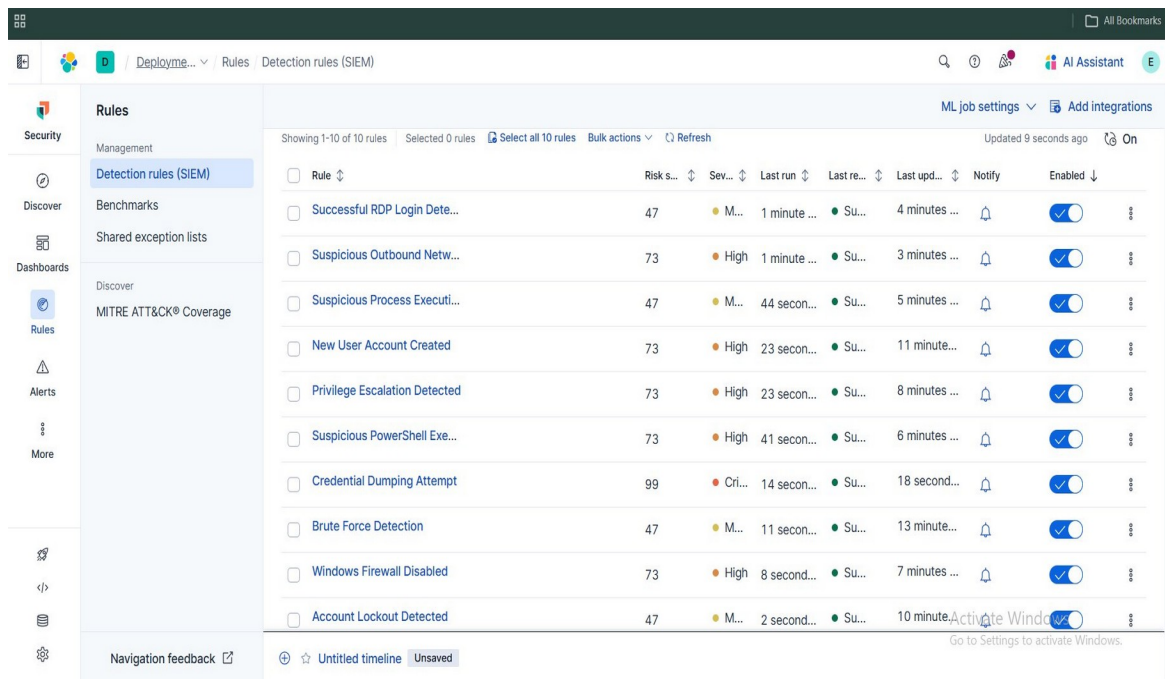
```
{
  "rule_name": "{{rule.name}}",
  "rule_id": "{{rule.id}}",
  "severity": "{{rule.severity}}",
  "timestamp": "{{date}}",
  "alert_id": "{{alert.id}}",
  "context": "{{context.message}}"
}
```

9. Elastic Detection Rules — All 8 Rules

Go to Kibana > Security > Rules > Detection Rules (SIEM) > Create New Rule for each rule below.

9.1 Rule Creation Process — Follow For Every Rule

- 93. Click Create New Rule
- 94. Select the Rule Type shown in the table below
- 95. Enter the exact KQL Query shown
- 96. For Threshold rules also set Group By and Threshold value
- 97. Click Continue and fill in Name, Severity, Risk Score
- 98. Schedule: Runs every 1 minute — Additional look-back 1 minute
- 99. Add Action: Webhook > N8N-SOC-Webhook > paste webhook body from Section 8.4
- 100. Click Create and Enable Rule



Kibana Detection Rules page showing all 8 rules created and enabled — blue toggle switches confirm all rules are active and running

Rule 1: Brute Force Detection

Field	Value
Rule Name	Brute Force Detection
Rule Type	Threshold
KQL Query	event.code: "4625"

Group By / Note	user.name — fires after 5 failures
Severity	Medium
Risk Score	47
MITRE Technique	T1110
Schedule	Every 1 minute Look-back 1 minute

Rule 2: New User Account Created

Field	Value
Rule Name	New User Account Created
Rule Type	Custom Query
KQL Query	event.code: "4720"
Group By / Note	—
Severity	High
Risk Score	73
MITRE Technique	T1136
Schedule	Every 1 minute Look-back 1 minute

Rule 3: Account Lockout Detected

Field	Value
Rule Name	Account Lockout Detected
Rule Type	Custom Query
KQL Query	event.code: "4740"
Group By / Note	—
Severity	Medium
Risk Score	47
MITRE Technique	T1110
Schedule	Every 1 minute Look-back 1 minute

Rule 4: Privilege Escalation Detected

Field	Value
Rule Name	Privilege Escalation Detected
Rule Type	Custom Query

KQL Query	event.code: "4732"
Group By / Note	—
Severity	High
Risk Score	73
MITRE Technique	T1078
Schedule	Every 1 minute Look-back 1 minute

Rule 5: Windows Firewall Disabled

Field	Value
Rule Name	Windows Firewall Disabled
Rule Type	Custom Query
KQL Query	event.code: "4950" or event.code: "4954"
Group By / Note	—
Severity	High
Risk Score	73
MITRE Technique	T1562.004
Schedule	Every 1 minute Look-back 1 minute

Rule 6: Suspicious PowerShell Execution

Field	Value
Rule Name	Suspicious PowerShell Execution
Rule Type	Custom Query
KQL Query	event.code: "4104" and winlog.event_data.ScriptBlockText: (*IEX* or *DownloadString* or *EncodedCommand* or *Mimikatz* or *bypass*)
Group By / Note	—
Severity	High
Risk Score	73
MITRE Technique	T1059.001
Schedule	Every 1 minute Look-back 1 minute

Rule 7: Suspicious Process Execution

Field	Value
-------	-------

Rule Name	Suspicious Process Execution
Rule Type	Custom Query
KQL Query	event.code: "4688" and winlog.event_data.NewProcessName: (*mshta* or *wscript* or *regsvr32* or *rundll32* or *certutil* or *cscript*)
Group By / Note	Suppress by host.name — 5 min
Severity	Medium
Risk Score	47
MITRE Technique	T1204
Schedule	Every 1 minute Look-back 1 minute

Rule 8: Suspicious Outbound Network

Field	Value
Rule Name	Suspicious Outbound Network
Rule Type	Custom Query
KQL Query	event.code: "5156" and winlog.event_data.DestPort: ("4444" or "1337" or "8888" or "9999")
Group By / Note	—
Severity	High
Risk Score	73
MITRE Technique	T1041
Schedule	Every 1 minute Look-back 1 minute

10. N8N Automation Workflow Setup

10.1 Create New Workflow

101. Login to your N8N Cloud account
102. Click New workflow button
103. Click on the title My workflow at the top and rename to: SOC-Automation

10.2 Node 1 — Webhook (Alert Receiver)

104. Click Add first step on the canvas
105. Search for Webhook and select it
106. HTTP Method: POST
107. Authentication: None
108. Click Production URL tab and copy the full URL
109. Save this URL in your credentials notepad — you need it for Section 8.3

Production URL: `https://yourname.app.n8n.cloud/webhook/xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx`

10.3 Node 2 — Groq AI (Alert Analysis)

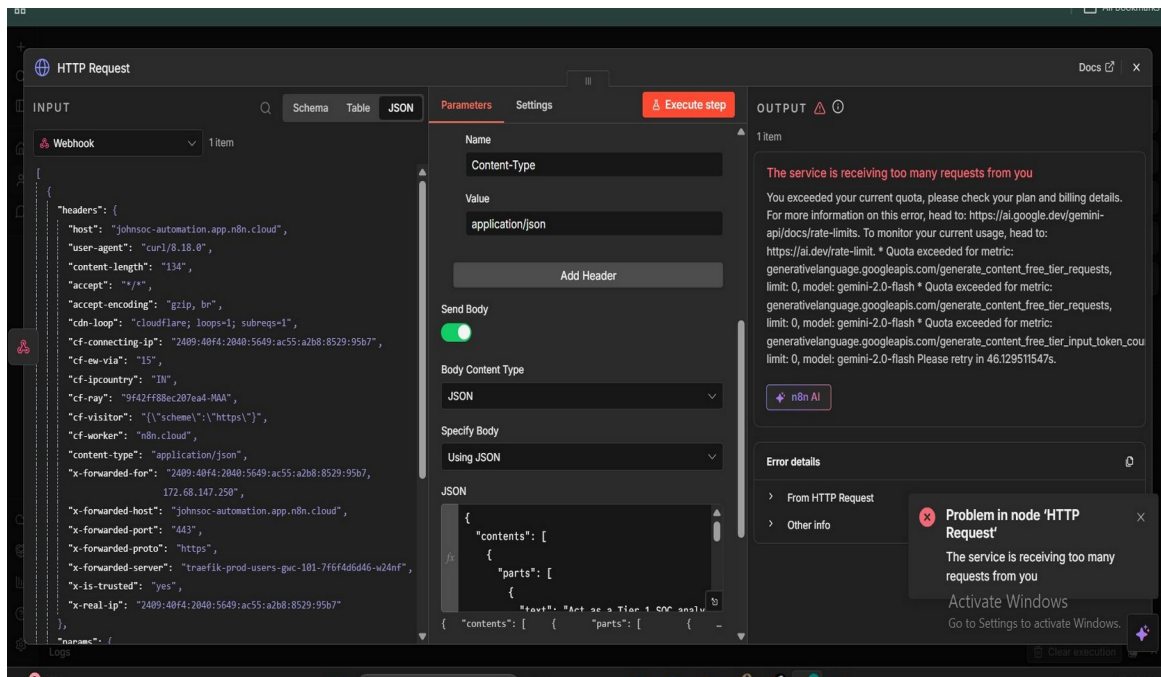
110. Click + after Webhook node > HTTP Request > rename to: Groq AI
111. Method: POST
112. URL: `https://api.groq.com/openai/v1/chat/completions`
113. Send Headers: ON — add these two headers:
 - Authorization: Bearer YOUR_GROQ_API_KEY
 - Content-Type: application/json
114. Send Body: ON — Body Content Type: JSON — paste this body:

```
{
  "model": "llama-3.3-70b-versatile",
  "messages": [
    {
      "role": "system",
      "content": "You are a Tier 1 SOC analyst. Analyze the security alert and provide:\n1. ALERT SUMMARY\n2. SEVERITY ASSESSMENT\n3. MITRE ATT&CK TECHNIQUE\n4. WHAT TO INVESTIGATE\n5. RECOMMENDED ACTIONS\nBe concise and professional."
    },
    {
      "role": "user",
      "content": "Analyze this SOC alert:\nRule: {{ $ ('Webhook').item.json.body.rule_name }}\nTime: {{ $ ('Webhook').item.json.body.timestamp }}\nContext: {{ $ ('Webhook').item.json.body.context || 'No context' }}"
    }
  ],
}
```

```

"temperature": 0.7,
"max_tokens": 1500
}

```

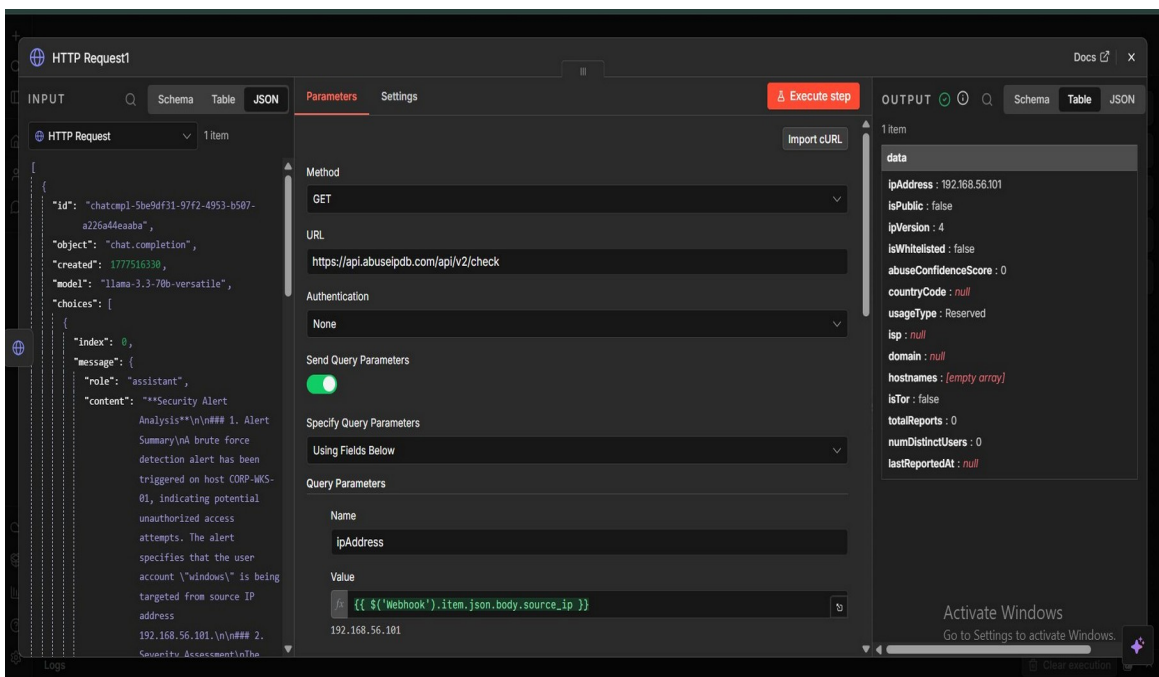


N8N Groq AI node configured and running — showing successful chat completion response from Llama 3.3 Groq model

11. AbuseIPDB Integration

11.1 Node 3 — AbuseIPDB Enrichment Node

115. Click + after Groq AI node > HTTP Request > rename to: AbuseIPDB Enrichment
116. Method: GET
117. URL: <https://api.abuseipdb.com/api/v2/check>
118. Send Query Parameters: ON — add these two:
 - Name: ipAddress | Value: {{ \$('Webhook').item.json.body.source_ip || '185.220.101.1' }}
 - Name: maxAgeInDays | Value: 90
119. Send Headers: ON — add these two:
 - Name: Key | Value: YOUR_ABUSEIPDB_API_KEY
 - Name: Accept | Value: application/json



N8N AbuseIPDB Enrichment node showing IP reputation data — abuse score 100, country DE, ISP Artikel10, Tor exit node confirmed

NOTE: The fallback IP 185.220.101.1 is used when alerts have no source IP. It is a known Tor exit node used to confirm the AbuseIPDB integration is working correctly.

12. Slack Integration

12.1 Node 4 — Slack Notification Node

120. Click + after AbuseIPDB Enrichment node
121. Search for Slack and select it
122. Action: Send a message
123. Click Connect to Slack and authorize your SOC-Automation workspace
124. Send Message To: Channel
125. Channel: Select alerts from dropdown
126. Message Type: Simple Text Message
127. Paste this complete message text:

```
🔥 SOC ALERT - {{ $('Webhook').item.json.body.rule_name }}
```

📋 ALERT DETAILS

- Rule: {{ \$('Webhook').item.json.body.rule_name }}
- Time: {{ \$('Webhook').item.json.body.timestamp }}

🤖 AI ANALYSIS

```
{{ $('Groq AI').item.json.choices[0].message.content }}
```

🔍 THREAT INTELLIGENCE

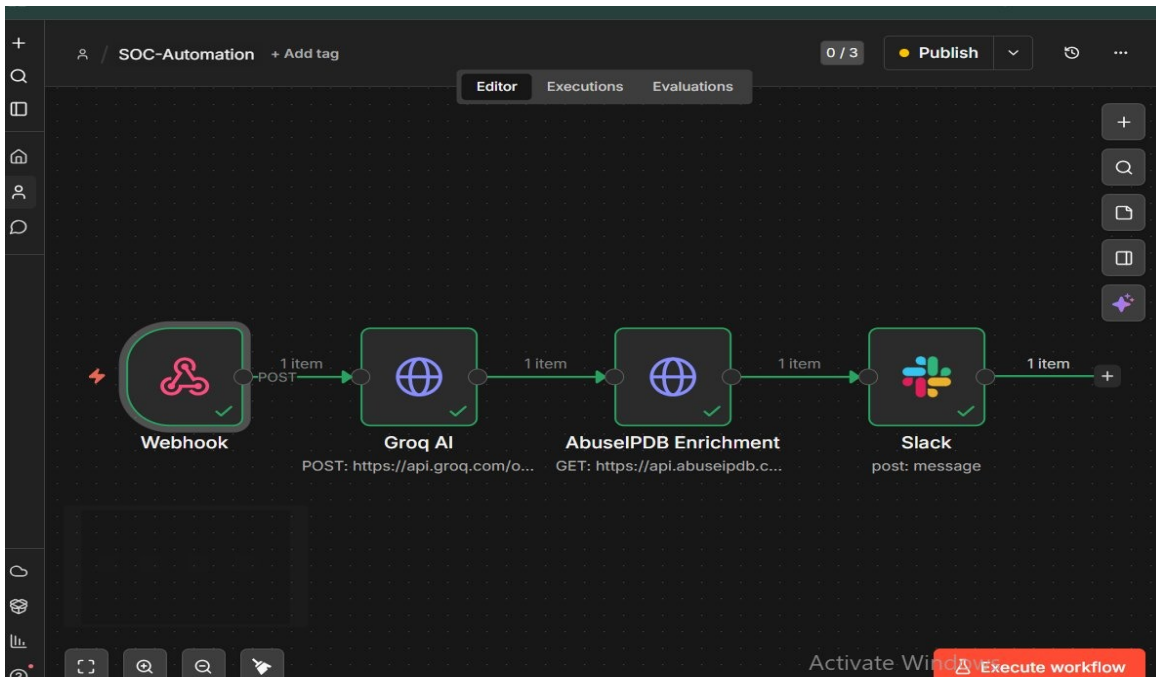
- Abuse Score: {{ \$('AbuseIPDB Enrichment').item.json.data.abuseConfidenceScore }}/100
- Country: {{ \$('AbuseIPDB Enrichment').item.json.data.countryCode || 'Unknown' }}
- ISP: {{ \$('AbuseIPDB Enrichment').item.json.data.isp || 'Unknown' }}
- Total Reports: {{ \$('AbuseIPDB Enrichment').item.json.data.totalReports }}
- Is Tor Exit Node: {{ \$('AbuseIPDB Enrichment').item.json.data.isTor }}

```
⚡ Powered by SOC-Automation
```

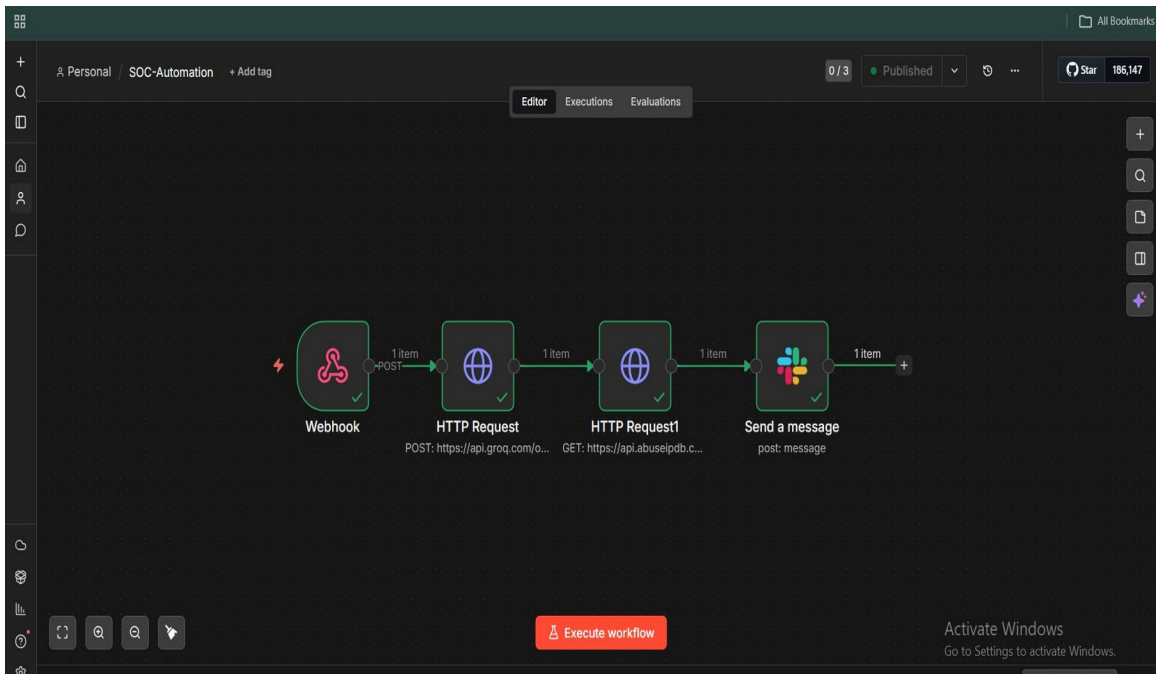
12.2 Publish Workflow

128. Close the Slack node panel
129. Verify canvas shows all 4 nodes connected: Webhook → Groq AI → AbuseIPDB Enrichment → Slack
130. Click Publish button top right

131. Version name: SOC-Automation-v1.0
132. Click Publish



N8N SOC-Automation workflow canvas showing all 4 nodes connected with arrows — Webhook, Groq AI, AbuseIPDB Enrichment, Slack — Published status top right



N8N published workflow showing green Published status — all nodes active and ready to receive alerts from Elastic Cloud

13. End-to-End Testing & Verification

Run these verification steps to confirm every component is working before running attack simulations.

13.1 Verify Winlogbeat is Shipping Logs

```
# In Kibana Discover – search: event.code: "4624" – Last 15 minutes
```

Expected: Windows login events visible from CORP-WKS-01

13.2 Verify N8N Webhook is Reachable

Test by sending a manual POST request to your webhook URL:

```
# On Kali Linux terminal:
curl -X POST https://yourname.app.n8n.cloud/webhook/YOUR-ID \
-H "Content-Type: application/json" \
-d '{"rule_name":"Test Alert","timestamp":"2026-01-01T00:00:00Z"}'
```

Expected: N8N Executions tab shows a new execution

13.3 Verify Full Pipeline

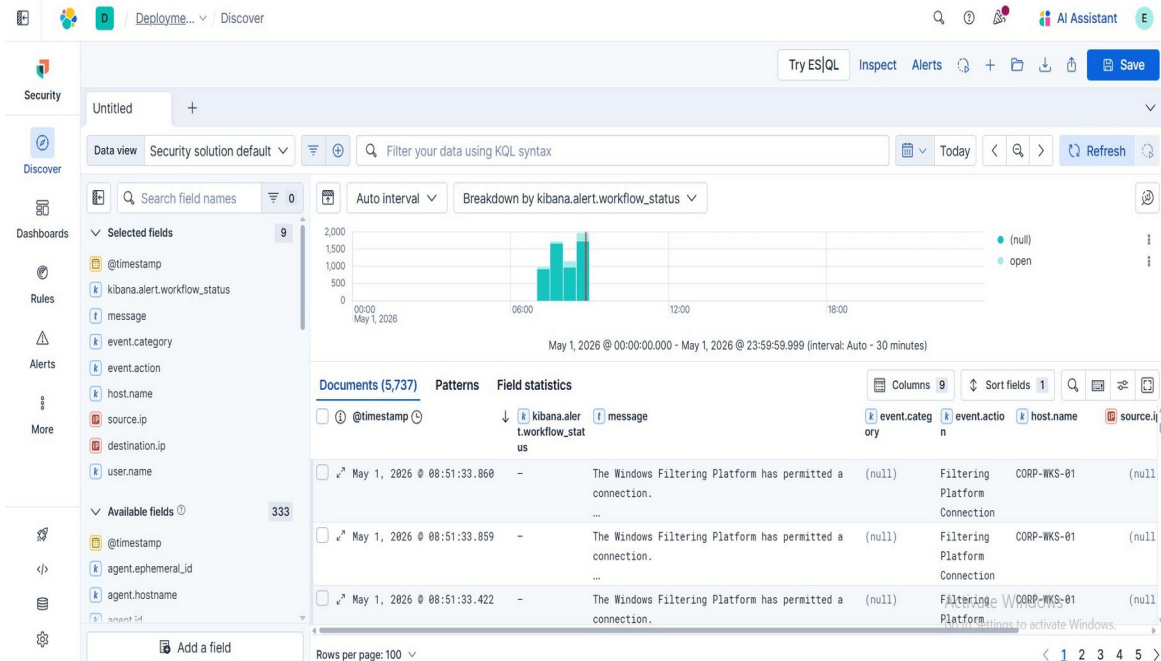
133. On Windows 10 VM PowerShell run:

```
Start-Process mshta.exe
```

134. Wait 2 minutes

135. Check Kibana Security > Alerts — should show Suspicious Process Execution

136. Check Slack #alerts — should show complete AI-analyzed alert



Kibana Security Alerts dashboard showing multiple detection rules firing — confirms complete pipeline is working end to end

14. Troubleshooting Guide

WINLOGBEAT ISSUES

Error	Cause	Fix
talk to server... ERROR 401	Wrong password in winlogbeat.yml	Reset elastic password at cloud.elastic.co and update yml
Script not digitally signed	PowerShell execution policy	Use: PowerShell -ExecutionPolicy Bypass - File install-service-winlogbeat.ps1
Config file error on start	YAML indentation wrong	Run: .\winlogbeat.exe test config -c .\winlogbeat.yml -e
No data in Kibana	Service stopped	Run: Start-Service winlogbeat
process.name not working	Wrong field name	Use winlog.event_data.NewProcessName instead
destination.port not found	Wrong field name	Use winlog.event_data.DestPort instead

N8N ISSUES

Error	Cause	Fix
Referenced node doesn't exist	Wrong node name in expression	Check exact node names on canvas — names are case sensitive
Webhook not receiving	Using Test URL not Production URL	Click Publish then use Production URL in Elastic connector
Groq API 429 rate limit	Too many requests	Wait 60 seconds — free tier has per-minute limits
AbuseIPDB invalid request	Empty IP in parameter	Add fallback: source_ip '185.220.101.1'
Slack channel not found	Channel name wrong	Verify channel is exactly: alerts (lowercase, no #)

ELASTIC CLOUD ISSUES

Issue	Cause	Fix
Rules not triggering	Query field name wrong	Test query in Discover first — verify correct field names
Webhook not sending	Wrong URL in connector	Update connector URL with N8N Production URL

Too many false positive alerts	Query too broad	Add specific field filters or enable alert suppression
No alerts in Kibana	Rule disabled	Go to Rules page and enable the rule — toggle must be blue
context.message empty	Elastic variable not supported	Use rule_name and timestamp only — they are reliable

WINDOWS EVENT ID QUICK REFERENCE

Event ID	Description	Triggered When
4624	Successful logon	Any successful Windows login
4625	Failed logon	Wrong password entered
4688	New process created	Any program or command starts
4720	User account created	net user /add command runs
4732	User added to group	net localgroup /add command runs
4740	Account locked out	Too many failed login attempts
4950	Firewall setting changed	netsh advfirewall command runs
5156	Network connection	Any outbound network connection made
4104	PowerShell script block	Any PowerShell command executed

15. Project Summary

You have successfully built a complete enterprise-grade SOC automation infrastructure using entirely free tools. This engineering setup demonstrates the ability to architect, configure and integrate multiple security tools into a working pipeline.

Component	Status	Verified By
VirtualBox + Windows 10 VM	Configured	ipconfig shows 192.168.56.102
Kali Linux VM	Configured	ip a shows 192.168.56.101
Windows Audit Policies	Enabled	auditpol /get returns Success and Failure
Winlogbeat Service	Running	Get-Service winlogbeat shows Running
Elastic Cloud Connection	Verified	test output shows talk to server... OK
Kibana Logs Flowing	Verified	Discover shows events from CORP-WKS-01
N8N Webhook Connector	Active	POST requests received in Executions tab
Groq AI Integration	Working	Chat completion response received
AbuseIPDB Integration	Working	IP reputation data returned
Slack Notifications	Working	Alerts received in #alerts channel
All 8 Detection Rules	Enabled	Rules page shows all rules active

Skills Demonstrated

Skill	Tool	Relevance
SIEM Administration	Elastic Cloud / Kibana	SOC Analyst L1/L2
Log Management	Winlogbeat 9.3.3	Security Engineer
Detection Engineering	KQL Query Language	Detection Engineer
Security Automation	N8N Workflow Engine	Security Automation Engineer
AI API Integration	Groq AI / Llama 3.3 70B	Modern SOC Analyst
Threat Intelligence	AbuseIPDB API	Threat Intelligence Analyst
Linux Administration	Kali Linux	Security Operations
Windows Security	Audit Policies / Event IDs	SOC Analyst
Virtualization	VirtualBox	Lab Engineering

